Product Analysis & Initiation Document (AID)

Project Name

Public Experience Analytics Platform (PEAP)

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1. Executive Summary

The Public Experience Analytics Platform (PEAP) is a crowd-driven analytics platform where users can register, verify their identity uniqueness, and provide ratings, votes, reviews, and experience-based feedback on products, services, organizations, and public entities used in day-to-day society.

The platform aggregates user sentiment and interaction data to generate real-time public analytical insights, trends, ratings, and visual reports.

The system aims to become a trusted public intelligence platform that reflects collective user experiences through transparent analytics and community-driven data.

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2. Business Vision

To build a globally accessible platform that helps society make informed decisions based on authentic collective experiences and public sentiment analysis.

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3. Business Objectives

1. Allow public users to create secure accounts and participate in evaluations.
2. Prevent duplicate or fake user identities.
3. Enable users to vote and provide experience-based feedback on entities.
4. Aggregate public opinion into meaningful analytical reports.
5. Display public analytics through graphs, charts, and trend indicators.
6. Establish a transparent reputation-based ecosystem for products, services, and organizations.
7. Create scalable APIs and backend services that support future mobile and web applications.

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4. Target Domains

The platform should support multiple categories including:

- Food items
- Consumer products
- Restaurants

- Government services
- Political parties
- Healthcare services
- Educational institutions
- Public transportation
- Telecommunication services
- Financial services
- E-commerce platforms
- Utility services
- Entertainment
- Social initiatives

The architecture should remain category-independent to allow future expansion.

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5. User Types

5.1 Guest User

Capabilities:

- Browse public analytics
- View public ratings
- View trend charts
- Search products/services
- Read public comments

Restrictions:

- Cannot vote
- Cannot submit reviews
- Cannot create analytics

5.2 Registered User

Capabilities:

- Create account
- Verify identity uniqueness
- Submit votes
- Submit ratings
- Add reviews/comments
- Report abuse/fraud
- Follow entities
- Receive notifications

5.3 Moderator

Capabilities:

- Review flagged content
- Remove abusive content
- Manage reports
- Moderate fake activity
- Approve sensitive submissions

5.4 System Administrator

Capabilities:

- Manage platform configuration
- Manage categories
- Manage analytics engines
- View audit logs
- Manage moderation policies
- User management

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6. Core Functional Requirements

6.1 User Registration & Identity Management

Features

- Email/password registration
- Social login support
- Multi-factor authentication (future)
- Identity uniqueness validation
- Device fingerprinting
- Anti-fraud validation
- Duplicate account detection

Business Rules

- One person should maintain only one active account.
- Suspicious duplicate accounts should be flagged.
- High-risk accounts may require additional verification.

6.2 Authentication & Authorization

Features

- JWT/OAuth2 authentication
- Session management
- Role-based access control (RBAC)
- Token refresh mechanism

6.3 Product/Entity Management

Features

- Create public entities
- Categorize entities
- Tagging system
- Approval workflow
- Entity search

Example Entities

- Soap brands
- Restaurants
- Telecom providers
- Political parties
- Public services

6.4 Voting System

Features

- Upvote/downvote
- Star rating
- Experience scoring
- Weighted trust scoring
- Vote history tracking

Rules

- One user vote per entity version.
- Users may update votes within configurable limits.
- Fraudulent voting patterns should be detected.

6.5 Review & Feedback System

Features

- Text reviews
- Experience tags
- Sentiment classification
- Review reactions
- Abuse reporting

Moderation

- AI-assisted moderation
- Spam filtering
- Toxic content detection

6.6 Public Analytics Engine

Features

- Real-time score calculation
- Trend analysis
- Category comparison
- Public sentiment analysis
- Demographic segmentation
- Time-based analytics

Example Analytics

- Most trusted food brands
- Public satisfaction trends
- Political sentiment changes
- Service quality rankings

6.7 Visualization & Reporting

Features

- Line charts
- Heat maps
- Pie charts
- Time-series graphs
- Trend indicators
- Public dashboards

Output Types

- Daily trends
- Weekly analytics

- Category leaderboards
- Public trust index

6.8 Search & Discovery

Features

- Full-text search
- Auto-suggestions
- Trending entities
- Filter by category
- Filter by location
- Filter by rating

6.9 Notification System

Features

- Vote milestone alerts
- Trending alerts
- Moderator notifications
- Abuse reports
- Followed entity updates

6.10 Fraud Detection & Trust Engine

Features

- Bot detection
- Fake review detection
- Suspicious voting pattern analysis
- Device/IP anomaly tracking
- Reputation scoring

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7. Non-Functional Requirements

7.1 Scalability

The platform must support:

- Millions of users
- High concurrent voting traffic
- Real-time analytics processing

- Horizontal scaling

Recommended architecture:

- Microservices
- Event-driven communication
- Distributed caching

7.2 Availability

Target:

- 99.9% uptime

Recommended:

- Kubernetes deployment
- Multi-zone deployment
- Auto-scaling

7.3 Security

Requirements:

- JWT security
- Encryption at rest
- HTTPS/TLS
- Rate limiting
- OWASP compliance
- Audit logging

7.4 Performance

Targets:

- API response < 300ms
- Search response < 1 second
- Analytics dashboard < 2 seconds

7.5 Compliance

Potential future compliance:

- GDPR
- Data retention policies

- User consent management

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8. Suggested High-Level Architecture

Backend Architecture

- Java 17+
- Spring Boot Microservices
- Kafka Event Streaming
- PostgreSQL
- Redis Cache
- Elasticsearch
- Kubernetes
- API Gateway

Suggested Services

Identity Service

Handles:

- authentication
- user uniqueness
- authorization

Product Service

Handles:

- entities
- categories
- metadata

Voting Service

Handles:

- votes
- ratings
- trust scoring

Review Service

Handles:

- reviews
- moderation
- sentiment

Analytics Service

Handles:

- aggregation
- trend generation
- analytics processing

Notification Service

Handles:

- alerts
- emails
- subscriptions

Fraud Detection Service

Handles:

- anomaly detection
- suspicious activity analysis

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9. Suggested Event-Driven Architecture

Example Kafka Topics:

- user-created
- vote-submitted
- review-submitted
- analytics-generated
- fraud-detected
- notification-requested

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10. Data Analytics Possibilities

Future advanced analytics:

- AI sentiment analysis
- Public trust prediction
- Regional behavior analysis
- Market trend forecasting
- Political trend prediction
- Consumer satisfaction indexing

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11. Future Enhancements

- Mobile applications
- AI recommendation engine
- Public API marketplace
- Blockchain verification
- Geo-based analytics
- Community reputation system
- AI-generated insight summaries

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12. Risks & Challenges

Risk	Description
Fake accounts	Users creating multiple identities
Manipulated voting	Coordinated influence attacks
Political sensitivity	High-risk public sentiment topics
Legal concerns	Defamation/privacy risks
Scalability	Massive analytics workloads
Data authenticity	Maintaining trusted insights

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13. Suggested MVP Scope

Initial MVP should include:

- User registration
- Login/authentication
- Product/entity creation
- Voting system
- Basic reviews
- Basic analytics dashboard
- Public graph display
- Fraud prevention basics

Exclude from MVP:

- AI sentiment analysis
- Advanced moderation AI
- Multi-region deployment
- Blockchain verification

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14. Success Metrics

Metric	Target
User registrations	100K+
Daily active users	20K+
Votes per day	500K+
Analytics response time	< 2 seconds
Fraud detection accuracy	> 90%
Platform uptime	99.9%

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15. Recommended Development Phases

Phase 1

Core Platform MVP

Phase 2

Analytics & Visualization Expansion

Phase 3

AI Sentiment & Trust Engine

Phase 4

Large-Scale Public Intelligence Platform

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16. Conclusion

The proposed platform aims to become a scalable public intelligence and experience analytics ecosystem powered by community participation and real-time analytical processing.

The platform’s core competitive advantage lies in:

- authentic crowd-driven insights
- trust-oriented analytics
- scalable event-driven architecture
- transparent public experience visualization

The backend architecture should be designed from the beginning to support high scalability, distributed processing, and future AI-driven analytical capabilities.